MILITARY SPECIFICATION
GYROSCOPE, UNIT 26 OF GUN FIRE CONTROL SYSTEM MK. 56

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers the procurement requirements for repair or modification of the gyroscope (Unit 26) of the Gun Fire Control System Mk. 56, plus replacement gyroscopes, Stable Element gyroscope, Mk. 43 (L.D. 480498) and Rate Gyroscope, Mk. 41 (L.D. 480499).

1.2 Classification. The gyroscope (Unit 26) covered by this specification shall consist of either gyroscope (SK 136333) or Mk. 42 (L.D. 480500), as specified in the contract or order.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS:

MIL-F-18870 - Fire Control, Optical and Associate Equipment Intended For Shipboard Use, The General Specification For Design and Procurement of

STANDARDS:

MIL-STD-129 - Marking for Shipment and Storage

DRAWINGS:

Bureau of Naval Weapons

BuOrd SK. 136333 - Gun Director Mk. 56, Mod 2, Gyroscope Unit
BuOrd LD. 480498 - Stable Element Gyroscope Mk. 43
BuOrd LD. 480499 - Rate Gyroscope Mk. 41
BuOrd LD. 480500 - Gyroscope Unit (Mk. 42) Unit 26 of Gun Fire Control System Mk. 56
BuOrd Dwg. 688146 -Metal Storage Case

PUBLICATIONS:

Bureau of Naval Weapons

OD 15223 - Gyroscope, Unit 26 of GFCS, Mk. 56, Test Procedure

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

FSC 1210

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3. REQUIREMENTS

3.1 General Requirements. Gyroscope (Unit 26) shall consist of a government furnished gyroscope (SK.136333) which shall be either repaired or modified in accordance with 3.1.1 or 3.1.2 as applicable.

3.1.1 Repair. When specified in the contract or order, the government furnished gyroscope shall be repaired and put in serviceable condition conforming to the requirements of Sk. 136333 and all documents listed thereon. The repaired gyroscope shall meet all the requirements of this specification except replaceability and interchangeability of the stable element gyroscope and the rate gyroscope.

3.1.2 Modification. When specified in the contract or order, the government furnished gyroscope shall be modified in accordance with the requirements of gyroscope, Mk. 42, LD. 480500 and all documents listed thereon. The modified gyroscope shall meet all the requirements of this specification.

3.1.3 Replacement Gyroscopes. When specified in the contract or order, replacement gyroscopes, Stable Element, Mk. 43, LD. 480498 and Rate gyroscope, Mk. 41, LD. 480499, shall be manufactured in accordance with the requirements of the applicable LD and all documents listed thereon. Replacement gyroscopes shall meet all the applicable requirements of this specification.

3.2 The applicable requirements of MIL-F-18870 shall apply when not specified otherwise by this specification or the applicable drawings.

3.3 Gyro Motor Performance.

3.3.1 Power Supply. The gyro motors shall operate on 115 volts, 400 cycles, 3-phase power. The current in any phase shall not exceed 200 milliamperes and any difference in the three phases shall not exceed 15 milliamperes.

3.3.2 Gyro Wheel Rotation. Gyro wheels shall rotate clockwise as seen from the dome end.

3.3.3 Gyro Wheel Speed. The motors shall attain synchronous speed after running no more than 15 minutes. The rate gyro wheel speed shall be 12,000 ± 1 RPM. The stable element gyro wheel speed shall be 12,000 ± 5 RPM.

3.3.4 Coast Time. Coast time of gyro motors shall be not less than 5 minutes nor greater than 9 minutes.

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3.4 Stable Element Gyro Requirements.

3.4.1 Preliminary Requirements.

3.4.1.1 Swing Test. With the erection gimbals secured, the gyro gimbals shall make 40 or more oscillations after being released from one stop in either the cross traverse or elevation axes.

3.4.1.2 Static Balance. The gyro shall be slightly bottom-heavy.

3.4.1.3 Dynamic Balance. With erecting magnet inverted, stable element gyro shall precess between 0.5 degree and 1.0 degree per minute when gyro is displaced to 15 degrees.

3.4.1.4 Stable Element Settling Points. The settling points shall fall within a circle of 0.6 degree radius, having its center at true vertical. Gyro shall settle within 5 minutes when displaced ± 5 degrees in elevation or cross traverse.

3.4.2 Stable Element Pickoffs.

3.4.2.1 The rotor winding resistance of the Zs and E pickoffs shall be between 300 and 450 ohms.

3.4.2.2 The resistance of the Zs stator winding shall be between 400 and 550 ohms.

3.4.2.3 The resistance of the E stator winding shall be between 650 and 850 ohms between each pair of terminals.

3.4.2.4 The excitation current of either pickoff rotor shall not exceed 22 milliamps with 6 volts 60 cps applied.

3.4.2.5 Zs stator excitation. The excitation current shall not exceed 18 milliamps with 6 volts 60 cps applied.

3.4.2.6 E stator excitation. The excitation current shall not exceed 12 milliamps with 6 volts 60 cps applied to any pair of terminals.

3.4.2.7 Pickoff Harmonics.

3.4.2.7.1 True Elevation. The minimum voltage across the E pickoff shall not exceed 35 millivolts ± 5 per cent at any elevation.

3.4.2.7.2 Cross Traverse. The break-away voltage (across terminals 2 and 4) of the Zs servo motor shall not exceed 15 volts.

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3.5 Rate Gyro Requirements.

3.5.1 Torque Motor Coil Resistance. The torque motor coil resistance shall be 1350 ohms ± 10 per cent.

3.5.2 Crossed-E Pickoff.

3.5.2.1 Crossed-E pickoff coils. The crossed-E pickoff coil resistance shall be 970 ohms ± 10 per cent.

3.5.2.2 Crossed-E excitation. The excitation current of the crossed-E pickoff assembly shall be 11.8 ± 0.6 milliamperes with 35 volts excitation applied.

3.5.2.3 Phase shift. The phase shift of the crossed-E pickoff shall not exceed 36 degrees.

3.5.2.4 Crossed-E gradient. The crossed-E pickoff gradient shall be between 325 and 490 millivolts (101 ± 21 millivolts per 1/2 degree displacement).

3.5.2.5 Crossed-E voltage. The null of the elevation and traverse outputs shall be 5 millivolts or less.

3.5.3 Vibration damper. The spring fingers should follow the weight for 1/64" to 1/32" displacement.

3.5.4 Rate gyro freedom. The limit stops shall be adjusted to allow at least ± 2 degrees motion between stops on both axes of its gimbal system.

3.5.5 Gimbal bearing friction. The algebraic difference between CW and CCW train torque motor currents at any elevation shall not exceed 30 microamperes. The algebraic difference between CW and CCW elevation torque motor currents at any elevation shall not exceed 45 microamperes.

3.5.6 Wander rate. The average torque motor current over an interval of two minutes shall not exceed ± 20 microamps at any elevation. (Accuracy of Measurement: ± 1 microampere.)

3.5.7 Crossed-E pickoff centering. With El at 90°, the difference in torque motor currents for CW and CCW rotation shall not exceed ± 30 microamps. (Accuracy of Measurement: ± 1 microampere.)

3.5.8 Magnetic axis alignment. When the crossed-E excitation is varied from 31.5 ± 0.2 volts to 38.5 ± 0.2 volts, the torque motor currents must not vary more than 20 microamps. (Accuracy of Measurement: ± 1 microampere.)

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3.5.9 Torque motor calibration.

3.5.9.1 Torque motor gradient. The torque motors shall be calibrated to produce a precession constant of 2.000 milliamps/deg/sec. On this basis, torque motor current shall indicate precession rate with an accuracy of ± 0.15% ± 0.175 milliradians/sec. for CW and CCW rotations (not the average of the two).

3.5.9.2 Train torque motor. When 60 milliamps of current is applied to the train torque motor, the elevation torque motor CW and CCW current spread shall not exceed 20 microamps. Standstill train and elevation torque motor currents shall not exceed ± 20 microamperes. The elevation torque motor CW and CCW currents shall not exceed ± 40 microamps. (Accuracy of Measurement: ± 1 microampere.)

3.5.9.3 Elevation torque motor. When 60 milliamps of current is applied to the elevation torque motor (E axis 0°, Zs axis 90°), the train torque motor CW and CCW current spread shall not exceed ± 50 microamps. The train CW and CCW currents shall be equal within ± 50 microamps to the train standstill current. The elevation torque motor current at standstill shall not exceed ± 40 microamps. (Accuracy of Measurement: ± 1 microampere.)

3.6 Scorsby Test.

3.6.1 Cross-traverse gearing and limit switches. When the Zs gearing is rotated between limits, there shall be no interference or binding. The limit switches shall be actuated at 89° to 90° and 270° to 271°.

3.6.2 Mechanical stops. The mechanical stops shall be encountered at 89° 30 minutes + 45 minutes, -30 minutes and 270° 30 minutes + 30 minutes, -45 minutes dial readings, respectively.

3.6.3 Backlash. With the cross-traverse zeroing pin inserted, the backlash at the Zs dials shall be not greater than two minutes.

3.6.4 Refined dial zero. The backlash shall be equally distributed about the dial zero.

3.6.5 Wiring and terminals. Wiring and terminals shall be in accordance with the following drawings: Original design gyro, Dwg. 588806; Mk. 42 gyro, Dwg. 1461662

3.6.6 Gyro nutation. A jitter in the gyro shall damp out within two minutes at any elevation.

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3.6.7 Elevation error. With elevation axis parallel to Scorsby axis at rolls of ± 10°, ± 15°, ± 20°, and ± 25° at approximately a 10-second period, elevation error shall not exceed ± 1.2°. (Accuracy of measurement: 20 minutes of angle.)

3.6.8 Cross-traverse error. At ± 10°, ± 15°, ± 20° and ± 25° rolls about the Zs axis at approximately 10-second periods, the cross-traverse error shall not exceed ± 0.5°. (Accuracy of measurement: 10 minutes of angle.)

3.6.9 Zs Pickoff Gradient. The Zs pickoff gradient shall not exceed 36 millivolts per degree ± 10 per cent.

3.6.10 Case Elevation error. With the gyro case set at 0°, 40°, 60° and 80°, in elevation, the limits listed below shall not be exceeded:

Case El	Zs Error	El Reading
0°	± 36 minutes	± 95 minutes
40°	± 1° 48 minutes	40° ± reading at 0° ± 95 minutes
60°	± 3° 28 minutes	60° ± reading at 0° ± 95 minutes
80°	± 12° 0 minutes	80° ± reading at 0° ± 95 minutes

3.7 Final turntable tests.

3.7.1 Zs Potentiometer.

3.7.1.1 Resistance. At 0° Zs the resistance shall be 0 ± 15 ohms.

3.7.1.2 Brush pressure. The wiper brush pressure shall be 30 ± 5 grams.

3.7.1.3 Collector ring brush pressure shall be 10-35 grams.

3.7.2 Zs Resolvers. The resolvers shall be zeroed so that R1R2 voltage equals R2R3 voltage within 0.3 volts.

3.7.3 Axis alignment.

3.7.3.1 The Zs axis shall be perpendicular to the E axis within 1 minute of angle.

3.7.3.2 The V-block center line shall be parallel to the Zs axis within 20 seconds of angle.

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3.8 Visual Examination.

3.8.1 All cover screws, terminal screws, and lock nuts shall be tight, and correct nameplate installed.

3.8.2 There shall be no interference of torque motor coils, wire clips, screws or cables when gimbal systems are rotated between stops.

3.8.3 Spiral connectors. When gyros are rotated to maximum stops, the spirals shall extend not more than 1/16 inch beyond the separators. The spirals shall not rub against the separators.

3.9 Hipot and megger. All circuits shall withstand a hipot test of 500 VAC for one minute. Immediately after hipot, insulation resistance shall be at least five megohms as read with a 500 VDC megger.

3.10 Workmanship. The equipment, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, machine screw assemblies, lock nut assemblies, freedom from metal chips and other foreign matter, welding, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Examination. Each gyro unit shall be examined for conformance to applicable drawings.

4.3 Test equipment and inspection facilities. Test equipment and inspection facilities shall be of sufficient accuracy and quality to permit performance of the required tests. The manufacturer or repair activity shall establish adequate calibration of test equipment to the satisfaction of the procuring activity or approved testing laboratory.

4.4 Acceptance tests. Each unit shall be tested in accordance with O.D. 15223 to assure that it meets the requirements specified in Section 3.

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4.5 Test procedures.

4.5.1 Gyro motor performance. When tested in accordance with O.D. 15223, the motors shall meet the requirements of Section 3.3.

4.5.2 Stable element gyro. The stable element gyro shall be tested in accordance with O.D. 15223 and meet the requirements of Section 3.4.

4.5.3 Rate gyro. The rate gyro shall be tested in accordance with O.D. 15223 and meet the requirements of Section 3.5.

4.5.4 Scorsby test. When tested in accordance with O.D. 15223, each gyroscope shall meet the requirements specified in Section 3.6.

4.5.5 Final turntable test. When tested in accordance with O.D. 15223, each gyroscope shall meet the requirements specified in Section 3.7.

4.5.6 Visual examination. The gyroscope unit shall be examined in accordance with O.D. 15223 and meet all requirements specified in Sections 3.8 and 3.10.

4.5.7 Hipot and megger. When tested in accordance with O.D. 15223, all circuits shall meet the requirements specified in Section 3.9.

5. PREPARATION FOR DELIVERY.

5.1 Packing.

5.1.1 Level A.

5.1.1.1 The gyroscope, Unit 26 or gyroscope Mk. 42 shall be packed in the reusable shipping container, BuOrd Dwg. No. 688146, in accordance with the instructions contained on the drawing.

5.1.1.2 Gyroscopes Mk. 41 and Mk. 43 shall be individually packed in a special reusable container as specified in the contract or order.

5.1.2 Level B (Not applicable).

5.1.3 Level C (Not applicable).

5.2 Marking. In addition to any special marking required by the contract or order, the shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129 and BuOrd Dwg. No. 688146. The marking shall include the serial number of the gyro contained therein.

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6. NOTES

6.1 Intended use. The gyro units covered by this specification will be used only in Gun Fire Control System, Mk. 56, all mods. The gyros are not interchangeable in any other system or equipment.

6.2 Ordering data. Procurement documents should contain the specific title, number and date of this specification and exceptions to this specification, applicable drawings and other documents. (See 1.2).

Notice. When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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